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Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

The Informational Cost of Spatial Prompts in SAM: A Controlled Study on Polyp Segmentation

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MOTIVATION & RESEARCH PROBLEM

- Segment Anything (SAM) [1] is often called a general segmenter, but its mask decoder expects a spatial prompt: a click or a box that tells it where and which object to segment.
- In colonoscopy, someone has to provide that prompt for every frame. Removing it is the step that would make SAM fully automatic.
- Earlier work generates prompts automatically [4, 5] and reports that SAM's chosen mask can trail its best candidate [3]. It rarely asks why SAM fails without a prompt, or whether SAM is still worth its cost once prompting is automated.

Research question: How much of SAM's polyp-segmentation quality comes from the spatial prompt? Where does it fail without one, and how much can an automatic replacement recover, at what cost?

OBJECTIVES / CONTRIBUTION

- **Measure prompt dependence** on the same images, from no prompt to oracle point and box.
- **Separate two failure modes:** no good mask candidate, versus a good candidate that SAM does not select.
- **Test automatic prompts against honest controls:** a fixed location prior and a U-Net trained from scratch.
- **Look beyond one score:** label budget, external transfer, false activation on polyp-free images, and CPU latency.

Worth discussing: should automatic SAM pipelines be judged only by Dice, or also by whether they can say "no polyp here"?

METHODOLOGY

- **Data:** Kvasir-SEG [2], 400 training / 100 development images, fixed IDs, 3 seeds. External, no tuning: CVC-ClinicDB [6] (612 frames, 29 sequences) and HyperKvasir cecum [7] (1,009 polyp-free images).
- **Model:** SAM ViT-B, fully frozen (FP32, 1024 px).
- **Comparator:** compact U-Net (487k parameters) trained from scratch on the same images.
- **Statistics:** Dice per image; 95% bootstrap CIs (by sequence for CVC-ClinicDB); paired differences.

[Method diagram: prompt sources → frozen SAM decoder · U-Net comparator · 3 evaluation sets]

RESULTS & DISCUSSION

Panels 1–3: development set (exploratory). Panel 4: frozen external protocol. Planned: abstention gate, held-out Kvasir test, SAM 2.1 / MedSAM.

1 · Prompt dependence
No prompt: Dice 0.009; GT box: 0.935. Probe → box + point recovers 87% of the gap (0.818); a fixed prior reaches only 0.358.

[Figure: Dice by prompt condition]

2 · No candidate vs. wrong pick
No prompt: even the best of 4 outputs reaches only 0.182. GT point: the best of 3 reaches 0.860, but SAM's IoU score picks 0.540.

[Figure: candidates vs. selection]

3 · Is SAM worth it?
Kvasir: SAM 0.818 vs. U-Net 0.827 (unresolved), ~309× slower on CPU. SAM leads under short training; a longer U-Net recipe closes the gap.

[Figure: label efficiency]

4 · Outside Kvasir
CVC-ClinicDB: SAM 0.459 vs. U-Net 0.340. Polyp-free HyperKvasir: 5.2% vs. 17.3% false area, yet both output a mask in ≥99.8% of images.

[Figure: CVC transfer + false area]

CONCLUSIONS

- Without a prompt, SAM lacks good candidates. With a point, it has them but picks poorly.
- Automatic localization (probe → box + point) recovers 87% of the prompt gap; a fixed prior does not.
- Against a U-Net from scratch, SAM is not better on Kvasir and is about 300× slower on CPU. It does start higher with little training and transfers better to CVC-ClinicDB.
- Limitations:
  - Choices were made on the development set.
  - Only one SAM size was tested.
  - The negatives are a class-level proxy.
  - Latency was measured on CPU only.
- Next: an abstention gate tested on grouped negative videos.

Take-home: The prompt tells SAM where to look. Automating that recovers most of the quality, but not the ability to say "there is no polyp here".

KEY REFERENCES

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CONTINUE THE CONVERSATION

Code, protocols and per-image results:
[QR CODE]
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